



ROV '25 CERNON MOTORS, ESCs and RELAYS

This document is written to clarify the chosen motors, ESCs and relays for the ROV “CERNON” and why did we choose them. We are going to differentiate between T200, T500, and M200 and W30 motors depending on performance, cost, reliability, and the requirements of the ROV.

ROBOTECH '25

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1.MOTORS

The type of motor in our recent ROV “ORCA” was **T100** however since it’s not available anymore, we had to search for other types of motors and decide which one of them is suitable.

There’re three types of motors we will compare which are **T200**, **T500**, and **M200** From **Blue Robotics** and **W30** from **CubeMars** with no locals because no local alternatives and the only local is a normal **BLDC** that have weaker performance, no water resistance, and less lifetime and cannot run the ROV.

First: COST

T200 is for 220\$, **T500** is for 750\$, **M200** is for 175-185\$ and is for **W30** 210\$ “with the ESC”, Therefore we’ve excluded T500 for its expensive price and it has triple the properties of T200 and its more than the ROV will ever need.

SECOND: PROPERITIES

The differences:

T200 , **M200** and **W30** differ in their length and diameter where:

T200’s length is 113 mm (about 4.45 in), and its diameter is 100 mm (about 3.94 in).

M200’s length is 73,6 mm (about 2.89 in) and its diameter is 40 mm (about 1.57 in).

W30’s length is 108 mm (about 4.25 in) and its diameter is 100 mm (about 3.94 in).

As noticed, **T200’s** and **W30’s** sizes are bigger than **M200’s**, which will be more suitable for the ROV to be able to move in the water smoothly.

They also differ in their weight in air and water as:

T200’s weight in air is 427 g and in water is 239 g.

M200’s weight in air is 282 g and in water is 217 g.

w30’s weight in air is 420 g and in water is 200 g.

The rotations per minute as:

At 16 volts and REV 1100 microsecond **T200** has 3465 RPM, **M200** has 2824 RPM and **W30** has 2989 RPM and at FWD 1900 microsecond **T200** has 3527 RPM, **M200** has 2751 RPM and **W30** has 3027 RPM which makes the thrust power of the **T200** stronger than **M200** and **W30** according to their data sheet.

Thermal management:

The cooling method of **T200** and **W30** is water- cooled (direct contact) while **M200** is air cooled

(indirect) Heat dissipation of **T200** and **W30** is excellent, due to water's high thermal conductivity while it is limited in **M200** as it relies on passive or external air flow.

Running time at full load of **T200** and **W30** is that they can run continuously underwater while **M200** needs cooling periods or active airflow to avoid overheating and it may require additional heat sinks or fans for high performance.

Components of **T200** are more compatible with ROV as it contains Buna-N which is resistant to oil, fuel and chemicals which the ROV is likely to be exposed to underwater.

The Power Consumption are the same for the T200 and the M200:

Full Throttle Current (Power) @ 12 V	17 Amps (205 Watts)
Full Throttle Current (Power) @ Nominal (16 V)	24 Amps (390 Watts)
Full Throttle Current (Power) @ Maximum (20 V)	32 Amps (645 Watts)

The Power Consumption for W30:

Full Throttle Current (Power) @ 16 V	12.5 Amps (200 Watts)
Full Throttle Current (Power) @ Nominal (19 V)	16.9 Amps (321 Watts)
Full Throttle Current (Power) @ Maximum (24V)	22.51 Amps (540 Watts)

The Performance:

Full Throttle FWD/REV Thrust at (16 V)" T200"	5.25 / 4.1 kg f
Full Throttle FWD/REV Thrust at (16 V)" M200"	5.63 / 2.81 kg f
Full Throttle FWD/REV Thrust at (16 V)" W30"	3.76 / 2.75 kg f

As both of **T200** and **M200** have the same power Consumption even if the **M200** slightly has a more performance in FWD, but it has a weak performance in the reverse so the **T200** has the perfect performance ratio between FWD and REV.

2.ECSs

There are two types of ESC “electronic speed controller” which are:

- I. Basic ESC (BESC 30-R3) and it is the type we are using with **T200** and **M200** thrusters in the ROV, and its price starts from 38\$ but no need to buy because the **T200** comes with its ESC 2-6S if we buy it from [MAKERS](#) in Egypt for 23,000 L.E.
- II. Basic ESC 500 which is used for **T500** thruster that we will not need.
- III. W30 ESC-8S-45A which is used for **W30** thruster, and its price starts from 38\$ but no need to buy because the **W30** comes with its ESC.

3.RELAYS

Normal Single relays are used at prices starting from 50 £ to 100 £ per piece and we will need 12 of them.

Resources:

Blue Robotics

[M200](#)

[T200](#)

CubeMars

[W30](#)